

Samuel Cabral Cruz

Senior Software Developer

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🗣️ English (Fluent) | French (Native)

Professional Summary

Senior Software Developer with 8+ years of experience designing and implementing scalable solutions across diverse technology stacks. Proven track record in leading technical initiatives, driving adoption of cutting-edge technologies, and supporting large development teams. Expert in microservices architecture, cloud infrastructure, observability, platform engineering, and developer productivity tooling. Specialized in modernizing legacy systems and refactoring complex codebases without disrupting production operations. Actively integrating AI-augmented development workflows to accelerate delivery and establish modern development practices. Strong background in both software engineering and actuarial science, combining analytical rigor with technical excellence.

Technical Skills

Architecture & Design: Domain-Driven Design (DDD), Event-Driven Architecture (EDA), Event Sourcing (ES), EventStorming, Test-Driven Development (TDD), System Design, Distributed Systems, Platform Engineering, Legacy Modernization, Incremental Refactoring

Programming Languages: Golang, TypeScript, Java, Python

Cloud & Infrastructure: AWS, Kubernetes, Docker, Pulumi, Terraform, Custom Resource Definitions (CRDs)

Backend: Microservices, GraphQL, REST APIs, Protobuf, Kafka

Frontend: React, Svelte, Storybook, Vite, Playwright

DevOps & CI/CD: GitHub Actions, GitLab CI, CircleCI

Monitoring & Observability: Prometheus, Grafana, Kibana, Splunk

Databases: DynamoDB, MongoDB, PostgreSQL, Elasticsearch, OpenSearch

AI-Augmented Development: Claude, Prompt engineering, AI-assisted code generation

Tools & Others: Unix/Linux, Bash, Mise, HTTPYac, Vim, Git, L^AT_EX

Professional Experience

Senior Software Developer

May 2024 – Present

Ticketmaster – Presto

Quebec, Canada

- Developed, maintained, and supported critical ticket and product delivery flow by SMS, ensuring reliable customer communication for ticketing operations
- Owned and stabilized shared infrastructure projects:
 - OpenSearch infrastructure supporting centralized logging and search capabilities for 7+ teams
 - Log forwarding pipeline enabling centralized log aggregation across the organization
- Brought the feature flag service to the Presto team and drove several improvements:
 - Optimized the deployment pipeline to reduce flag propagation time
 - Created React hooks to streamline feature flag integration for frontend projects
 - Integrated the service into the gateway so backend services automatically receive active feature flags and adapt their behavior in sync with the frontend

- Added context extrapolation to support dynamic, entity-aware flag evaluation
- Added header-based flag override support to enable developers to test feature flag behavior without modifying flag definitions
- Designed and delivered a centralized configuration service adopted across 5 stakeholder groups, featuring a Kafka-based async propagation pipeline with eventual consistency and sub-2-minute SLA, JSON Schema validation, auditing, and a permission model — exposed through 3 distinct interfaces (REST for backend consumers, GraphQL for frontend, web UI for support operations)
- Led the sunset of a shared log forwarding project, collaborating with integrators to migrate their logging infrastructure to Splunk while ensuring zero data loss and minimal disruption
- Provided technical guidance and support to integration teams during the migration, ensuring smooth transition and knowledge transfer
- Participated in on-call rotation for production systems, ensuring high availability of critical delivery services

Technologies: AWS, Claude, Docker, GitLab CI, Grafana, Java, Kafka, Kubernetes, Mise, OpenSearch, Playwright, PostgreSQL, Prometheus, Pulumi, Python, React, Splunk, TypeScript, Vite

Senior Software Developer

August 2023 – May 2024

Ticketmaster – TM1 Core Platform

Quebec, Canada

- Owned and evolved a micro-UI integration platform serving 150+ developers across multiple teams, refactoring dependency injection flows to improve modularity and performance
- Drove adoption of cutting-edge technologies and established coding standards, influencing architectural decisions across development teams
- Developed and maintained developer productivity tools eliminating critical roadblocks:
 - Feature flag service for controlled rollouts, adopted by 25+ teams
 - Kubernetes operator with Custom Resource Definitions (CRDs) to manage micro-frontend component inventory and automate import map versioning
 - Pulumi configuration resolution library for infrastructure as code
 - Slack bot in Go to verify inter-service access by region, enabling developers to troubleshoot connectivity issues directly from Slack
- Provided technical support and mentorship to 150+ developers, facilitating platform integration and tool adoption
- Monitored service usage and adoption metrics using Prometheus and Grafana, informing product decisions
- Participated in on-call rotation for development and production environments, ensuring high availability
- Established and promoted development best practices and technical standards across the organization

Technologies: AWS, Docker, GitLab CI, Golang, Grafana, Java, Kubernetes, CRDs, Mise, OpenSearch, Playwright, Prometheus, Pulumi, React, TypeScript, Vite

Software Developer

June 2020 – August 2023

Ingeno

Quebec, Canada

- Delivered software solutions as an external consultant across 6+ client engagements in diverse industries, with roughly half involving greenfield development from the ground up
- Contributed to diverse development projects across multiple domains:
 - Marketing analytics application for event management platform
 - POS integration facade for online food ordering and delivery platform
 - Smart objects lifecycle management platform
 - Commercial and residential condos building service management platform
 - Surgeries scheduling optimization UI

- Core domain for advertisement placement optimization company
- Conducted technical audits and provided recommendations for architectural improvements
- Managed client interactions, translating business requirements into technical solutions
- Created and maintained project configuration templates to accelerate team onboarding
- Mentored bachelor students in their capstone projects, providing technical guidance
- Participated in recruiting process, conducting technical interviews and evaluating candidates

Technologies: .NET, AWS, AWS CDK, AWS Lambda, Chakra UI, CircleCI, DynamoDB, Elasticsearch, GitHub Actions, GitLab CI, Google Cloud, Hadoop, Helm, Java, Jest, Kibana, Kubernetes, MongoDB, Node.js, Prometheus, React, Storybook, Terraform, TypeScript, Vue.js

Software Developer

May 2019 – June 2020

DashThis

Quebec, Canada

- Maintained dashboard creation platform for tracking online advertising campaign performance
- Participated in product design workshops, contributing technical insights to feature planning
- Ensured quality of weekly releases through manual testing and QA processes
- Instrumented application to analyze user behavior using Google Analytics and Hotjar
- Managed integration of different codebase versions within a monolithic architecture
- Refactored legacy code to improve maintainability and performance

Technologies: .NET, Azure, C#, Google Analytics, GraphQL, Hotjar, Jest, Octopus Deploy, React, Storybook, Styled Components, TeamCity

Course Assistant – GLO-4002

May 2019 – May 2023

Laval University – Software Quality and Metrics

Quebec, Canada

- Original contributor of RaftmanCG, a platform for automated university project delivery and correction
- Corrected semester projects and exams
- Produced educational content for the course
- Answered students' questions and provided academic support

Technologies: Docker, Digital Ocean, Factory Bot, PostgreSQL, RuboCop, RSpec, Ruby on Rails, AWS S3, minitest

Course Assistant – GLO-2003

January 2020 – May 2020

Laval University – Introduction to Software Engineering Processes

Quebec, Canada

- Prepared and animated lab sessions
- Prepared and corrected quizzes and semester projects
- Wrote Karate/Cucumber test suites used in course material
- Answered students' questions and provided academic support

Technologies: Cucumber, GitHub Actions, Java, Karate, Node.js, Pandoc, Python, TypeScript

Software Developer – Intern

January 2019 – May 2019

XpertSea

Quebec, Canada

- Refactored business administration platform to improve code quality and user experience
- Modified Python backend to support new features and fix bugs

- Migrated functionalities from Ember.js to Polymer (Web Components + Shadow DOM)

Technologies: Ember.js, Jest, Polymer, Python, Semaphore CI, Sinon, Web Components

Actuarial Analyst – Property Ratemaking

May 2013 – December 2017

Desjardins General Insurance Group (DGIG)

Quebec, Canada

- Contributed to integration of State Farm’s Canadian book of business
- Developed KPI monitoring dashboard using Excel/VBA for conversion solution tracking
- Established budget assumptions for 2017 planning cycle
- Wrote scripts to identify, extract, produce, and validate pricing system information
- Reviewed segmentation and pricing models, ensuring actuarial accuracy
- Performed bulk extraction from indemnification agent notes using Natural Language Processing (NLP)
- Redesigned data extraction pipeline for fraud model development
- Created SAS scripts documentation generator using XSD, XML, and XSL

Technologies: Excel, SAS, SQL, VBA, XSD, XML, XSL

Education

Bachelor’s Degree in Software Engineering (B. Eng.)

January 2017 – May 2020

Laval University

Quebec, Canada

GPA: 4.22/4.33 | Minors: International and Distinction

Master’s Degree in Computer Science

January 2018 – June 2018

Norwegian University of Science and Technology (NTNU)

Trondheim, Norway

Student exchange program

Bachelor’s Degree in Actuarial Science (B. Sc.)

September 2012 – May 2015

Laval University

Quebec, Canada

GPA: 3.96/4.33

Certifications

Cloud & DevOps

- **AWS Certified Developer – Associate** – Amazon Web Services (2023)
- **Site Reliability Engineering Foundation** – DevOps Institute (2023)
- **DevOps Foundation** – DevOps Institute (2023)
- **AWS Certified Cloud Practitioner** – Amazon Web Services (2021)

Relevant Skills: Cloud architecture, Infrastructure as Code (IaC), Site Reliability Engineering (SRE), CI/CD pipelines, incident management, Service Level Objectives (SLOs), Service Level Agreements (SLAs), Service Level Indicators (SLIs)

Agile & Scrum

- **Professional Scrum Master™ I (PSM I)** – Scrum.org (2023)
- **Professional Scrum Developer™ I (PSD I)** – Scrum.org (2023)
- **Professional Scrum Product Owner™ I (PSPO I)** – Scrum.org (2023)
- **Professional Scrum with Kanban™ I (PSK I)** – Scrum.org (2023)

Relevant Skills: Scrum, Kanban, agile methodologies, sprint planning, backlog management, cross-functional team collaboration

Actuarial Science & Financial Risk

- **Casualty Actuarial Society (CAS) – 10 Professional Exams Passed (2013-2016):**
 - **CAS CP:** Course on Professionalism – Ethics and professional standards (2016)
 - **CAS 5:** Ratemaking and Claim Liabilities – Pricing models and loss reserving (2015)
 - **CAS C2:** Accounting, Law, and Regulation – Financial reporting and compliance (2015)
 - **CAS ST:** Stochastic Processes – Time series, Markov chains, Monte Carlo simulation (2014)
 - **CAS LC:** Life Contingencies – Survival models and life insurance mathematics (2014)
 - **CAS C1:** Risk Management – Enterprise risk management frameworks (2014)
 - **CAS 4:** Actuarial Models – Statistical modeling and predictive analytics (2014)
 - **CAS 3F:** Financial Economics – Portfolio theory, asset pricing, capital markets (2013)
 - **CAS 2:** Financial Mathematics – Time value of money, derivatives, financial instruments (2013)
 - **CAS 1:** Probability – Statistical foundations and probability theory (2013)

Relevant Skills: Financial modeling, risk analysis, stochastic modeling, statistical inference, regulatory compliance

Selected Projects

Going – Go Libraries Mono-repository

2024

Developed comprehensive Go libraries leveraging generics to provide common utilities including dependency injection, mocks, test utilities, error handling, and expectations builder.

Available at: github.com/SamuelCabralCruz/going

Commit Bucket Deploy (CBD)

2021

Created GitHub Action to deploy static websites within pull request lifecycle, enabling preview deployments for code reviews. Built with GitHub Actions, Bash, AWS CDK, and TypeScript.

Available at: github.com/SamuelCabralCruz/cbd

Free Code Coverage

2021

Developed GitHub Action to enforce code coverage maintenance on pull requests. Published to GitHub Marketplace.

Available at: github.com/marketplace/actions/free-code-coverage

Unresolved Review Threads

2021

Built GitHub Action to block pull requests with unresolved review threads. Implemented with TypeScript and Jest. Published to GitHub Marketplace.

Available at: github.com/marketplace/actions/unresolved-review-threads

RaftmanCG – University Projects Delivery Platform

2019

Architected and developed comprehensive platform for university project management featuring: project statements publication, delivery management, automated correction, team management, deliveries archive, and reviews/results publication. Built with Ruby on Rails, Active Record, RSpec, and Docker. Deployed on Digital Ocean.